TASK 4: TYRE WEAR ROOT-CAUSE AND GEOMETRY INVESTIGATION

Autocare Technician Level 2 | Block 8 | v2.1

TIME	EVIDENCE	PROGRAMME POINT
60 minutes	**Practical | Knowledge | Documentation**	**Weeks 71-80**

Your task

A vehicle has excessive or uneven tyre wear. You must inspect the tyres and related steering/suspension geometry, record evidence and identify the likely root cause before recommending work.

What you must do

1. Prepare the vehicle and complete tyre/alignment pre-checks.
2. Record tyre wear patterns and pressures.
3. Measure or interpret alignment/geometry data as instructed.
4. Identify likely root cause and recommended action.
5. Record findings and reinstate the workstation.

What good evidence looks like

SAFE SETUP
Suitable PPE, a prepared work area and controlled vehicle or equipment.

CORRECT METHOD
A logical sequence using the relevant procedure and technical information.

ACCURATE EVIDENCE
Actual readings, findings and actions recorded clearly and precisely.

SOUND DECISIONS
Appropriate recommendations, referral or authorisation supported by evidence.

You will need

- Light vehicle
- Tread depth gauge and tyre pressure gauge
- Alignment equipment or sample printouts
- Manufacturer geometry data
- PPE and Apprentice Assessment Record

SOW ALIGNMENT

Block 8 - Advanced Diagnostics and Workshop Management
Duties: 1-10 KSB evidence focus: K15-K21, S10-S15, B1, B2
Task reference: Block_8_Task_4_Tyre_Wear_Root_Cause

TASK 4: TYRE WEAR ROOT-CAUSE AND GEOMETRY INVESTIGATION

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

TASK COMPLETED AT

○ **APPRENTICE'S WORKPLACE**

○ **TRAINING CENTRE**

Vehicle make

Model

Registration

Mileage

Engine size / type

VIN / chassis number

Measurements and technical checks

Tyre wear pattern		NS/F tread depth	
OS/F tread depth		Front toe measured	
Front toe specification		Camber measured	
Tyre pressure		Suspected root cause	

Special tools, equipment or calibration requirements

Task evidence

PPE and precautions used

Safety-critical findings and immediate action

Work completed, findings and recommendations

Knowledge check

1. What tyre wear pattern can incorrect toe create?

2. Why must suspension condition be checked before alignment?

Knowledge check - continued

3. (a) How can tyre pressure affect wear patterns? (b) Why is rear alignment or thrust angle important?

4. How would you explain the benefits of correct geometry?

5. (a) What evidence confirms root cause rather than symptom? (b) What recommendation should be made if a component is worn?

Assessor/Workplace Mentor observation

- [ ] Inspect tyre wear patterns and related components.
- [ ] Record geometry or alignment measurements where available.
- [ ] Identify likely root causes using evidence.
- [ ] Prioritise repairs, adjustment or referral.
- [ ] Explain the customer benefit of correct geometry.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- ○ **PASS**
- ○ **REFER**

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

PIN (if applicable)

Apprentice signature

Assessor/Workplace Mentor signature

Date

ADAS K17 — repair checks and calibration

For tyre wear and geometry investigation, use manufacturer information for a vehicle or a clearly labelled case study:
1. Explain whether the work could affect ADAS and identify the required checks.
2. Explain any relevant static and dynamic calibration methods and equipment, or why specialist advice is needed.
3. Record your information source and referral decision. Do not assume every repair requires calibration or carry out work beyond your authorised role.

Your answer

ADAS K18 — driver assistance and responsibilities

Use a vehicle or clearly labelled case study from the tyre/geometry task. Identify a related driver-assistance feature and explain whether it warns the driver or actively assists. Explain the driver's responsibilities and, within the levels of driving automation, identify the level supported by the information provided. Name the information source; do not assume features or capabilities that the example does not show.

Your answer

BLOCK 8 | TASK 4

Autocare Level 2

Answer continuation

Use this page when an answer will not fit visibly in the original box. Reference the question and write "See continuation page" in the original answer. Keep all text visible for printing; add further numbered copies as needed.

Apprentice name

Additional page reference

Question / section reference

Continued answer

Question / section reference

Continued answer